**Cambridge International Examinations**
Cambridge International General Certificate of Secondary Education

| | |
|---|---|
| CANDIDATE NAME | |
| CENTRE NUMBER | |
| CANDIDATE NUMBER | |

*6533820761*

**CHEMISTRY** **0620/22**

Paper 2 **February/March 2015**

**1 hour 15 minutes**

Candidates answer on the Question Paper.

No Additional Materials are required.

**READ THESE INSTRUCTIONS FIRST**

Write your Centre number, candidate number and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
Electronic calculators may be used.
A copy of the Periodic Table is printed on page 16.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

This document consists of **14** printed pages and **2** blank pages.

IB15 03_0620_22/3RP
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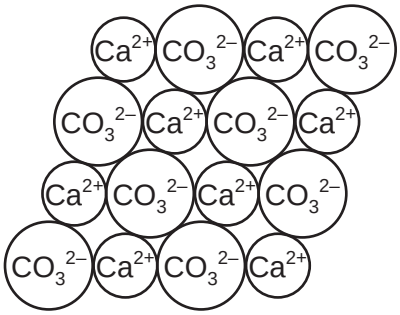
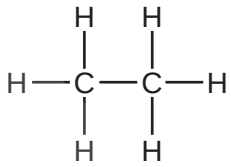
**1** The diagram shows the structures of some substances containing carbon.

**A**

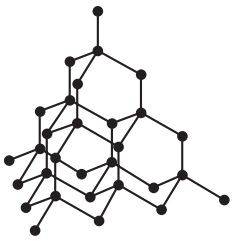



**B**

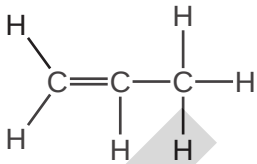
**C**



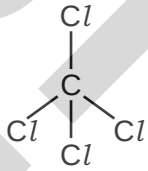
**D**



**E**



**F**



Answer the following questions about these substances.
Each substance may be used once, more than once or not at all.

**(a)** Which substance, **A**, **B**, **C**, **D**, **E** or **F**

**(i)** is a saturated hydrocarbon, .....

**(ii)** has an ionic structure, .....

**(iii)** is a product of respiration, .....

**(iv)** is in the same homologous series as methane, .....

**(v)** is used for cutting? .....

[5]

**(b)** Substance **D** is an element.

Explain why substance **D** is an element.

..... [1]

[Total: 6]

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**2** Some properties of the halogens are shown in the table.

| halogen | boiling point / °C | state at room temperature and pressure |
|---|---|---|
| fluorine | –188 | |
| chlorine | –35 | gas |
| bromine | +59 | liquid |
| iodine | +184 | solid |
| astatine | | solid |

**(a)** Use the information in the table to deduce

**(i)** the boiling point of astatine,

..... [1]

**(ii)** the state of fluorine at room temperature and pressure.

..... [1]

**(b)** When chlorine reacts with aqueous potassium iodide, the solution turns brown.

**(i)** Write a word equation for this reaction.

..... [2]

**(ii)** Explain why iodine does not react with aqueous potassium chloride.

.....

..... [1]

**(c)** When sodium reacts with iodine, energy is released.

**(i)** What is the name given to a reaction which releases energy?

..... [1]

**(ii)** Explain what happens in terms of electron transfer when a sodium atom reacts with an iodine atom.

.....

..... [2]

[Total: 8]

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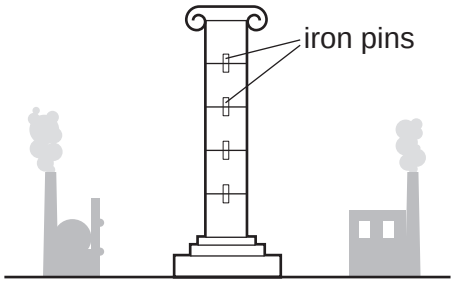
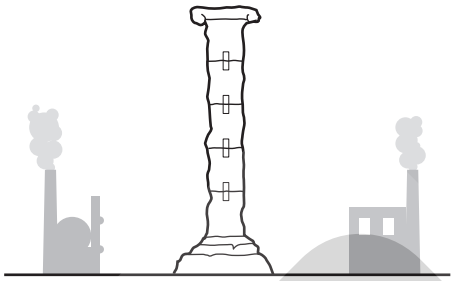
**3** The diagram shows a limestone column in an industrial town. Limestone is largely calcium carbonate.




column when first built



the same column after 40 years

**(a)** Describe and explain the changes to the column over 40 years.
In your answer refer to

- the change to the limestone,
- the name of a pollutant causing this change,
- the chemistry involved in this change.

.....

.....

.....

.....

.....

.....

..... [4]

**(b)** The sections of the column are joined with iron pins which rust when exposed to the atmosphere.

Describe **two** methods of rust prevention and explain how they prevent rusting.

.....

.....

.....

..... [3]

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**(c)** Iron is a transition element.

Give **two** properties of transition elements that make them different from non-transition metals such as magnesium.

.....

..... [2]

**(d)** An isotope of iron has 58 nucleons.

Complete the table to show

- the number of electrons and neutrons in this isotope of iron,
- the relative charges on each particle.

| particle | number of each particle present | relative charge on the particle |
|---|---|---|
| electron | | |
| neutron | | no charge |
| proton | 26 | |

[4]

**(e)** Iron reacts with hydrochloric acid to form iron(II) chloride and a gas which 'pops' with a lighted splint.

Complete the symbol equation for this reaction.

$$Fe + \ldots\ldots HCl \rightarrow FeCl_2 + \ldots\ldots\ldots\ldots$$

[2]

[Total: 15]

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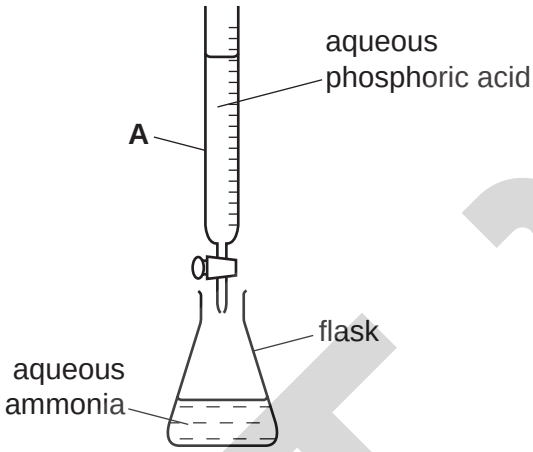
**4** Ammonium phosphate, $(NH_4)_3PO_4$, is a fertiliser.

**(a)** Which **two** elements in ammonium phosphate are important for plant growth?

..... and ..... [1]

**(b)** Aqueous ammonium phosphate can be made in the laboratory by reacting aqueous ammonia with aqueous phosphoric acid.




**(i)** State the name of the piece of apparatus labelled **A**.

..... [1]

**(ii)** Suggest the pH value of aqueous phosphoric acid.

..... [1]

**(iii)** Describe how the pH of the mixture in the flask changes as the acid is added.

..... [1]

**(iv)** Which **one** of the following best describes the reaction of aqueous ammonia with aqueous phosphoric acid?
Put a ring around the correct answer.

**combustion** **decomposition** **neutralisation** **reduction**

[1]

**(c)** When sodium hydroxide is added to ammonium phosphate, ammonia is released.

Complete the symbol equation for this reaction.

$$(NH_4)_3PO_4 + 3NaOH \rightarrow Na_3PO_4 + .....NH_3 + 3 .....$$

[2]

[Total: 7]

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**5** The table shows the concentration of some ions present in a sample of seawater.

| name of ion | formula of ion | concentration in g / dm$^3$ |
|---|---|---|
| bromide | $Br^-$ | 0.06 |
| calcium | $Ca^{2+}$ | 0.30 |
| chloride | $Cl^-$ | 20.00 |
| | $I^-$ | 0.04 |
| magnesium | $Mg^{2+}$ | 1.00 |
| potassium | $K^+$ | 0.50 |
| sodium | $Na^+$ | 11.00 |
| sulfate | $SO_4^{2-}$ | 0.80 |

**(a)** **(i)** Which positive ion in the table has the lowest concentration?

..... [1]

**(ii)** Give the name of the ion with the formula $I^-$.

..... [1]

**(iii)** Which **two** ions in the table are formed from elements in Group II of the Periodic Table?

..... and ..... [1]

**(iv)** Give the names of **two** ions in the table which move towards the anode (positive electrode) when a sample of this seawater is electrolysed.

..... and ..... [2]

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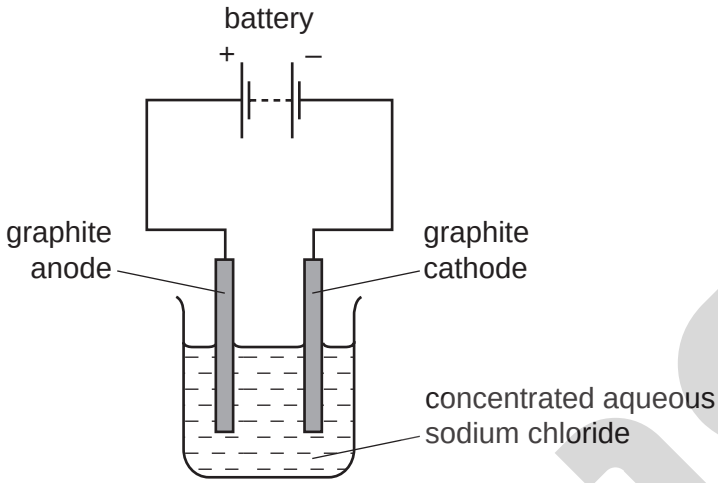
**(b)** Sodium chloride can be extracted from seawater.
Concentrated aqueous sodium chloride is electrolysed using the apparatus shown.




**(i)** Suggest why the anode and cathode are made of graphite.

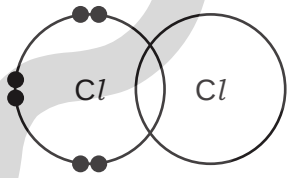
..... [1]

**(ii)** Give the name of the product formed at the cathode (negative electrode).

..... [1]

**(iii)** Chlorine is formed at the anode.

Complete the electronic structure of a chlorine molecule. Show only the outer shell electrons.




[2]

**(c)** Molten magnesium bromide is electrolysed.

Predict the products at the anode (positive electrode) and cathode (negative electrode).

anode .....

cathode .....
[2]

[Total: 11]

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**6** Zinc oxide is used for making baby soap and cream for treating sunburn.

**(a)** Suggest why the zinc oxide used for these purposes needs to be pure.

..... [1]

**(b)** Zinc oxide can be reduced by carbon. Carbon monoxide is one of the products.

**(i)** What is the meaning of the term *reduction*?

..... [1]

**(ii)** Write a word equation for the reaction of zinc oxide with carbon.

..... [1]

**(iii)** Explain why, in the laboratory, the reaction should be carried out in a fume cupboard.

..... [1]

**(c)** The table shows how easy it is to reduce various metal oxides by heating with carbon.

| metal oxide | ease of reduction with carbon |
|---|---|
| lead oxide | easily reduced at 300 °C |
| magnesium oxide | not reduced at 900 °C |
| nickel oxide | easily reduced at 500 °C |
| zinc oxide | fairly easily reduced at 900 °C |

Use the information in the table to put the metals in order of their reactivity.

least reactive ⟶ most reactive

| | | | |
|---|---|---|---|
| | | | |

[2]

**(d)** Zinc oxide reacts with sulfuric acid.

Complete the word equation for this reaction.

zinc oxide + sulfuric acid → zinc sulfate + .....

[1]

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**(e)** Pure dry crystals of zinc sulfate can be made by the reaction of dilute sulfuric acid with excess zinc.

**(i)** How is excess zinc removed from the reaction mixture?

..... [1]

**(ii)** Describe how you would obtain pure dry crystals of zinc sulfate from an aqueous solution of zinc sulfate.

.....

.....

.....

..... [3]

**(iii)** Zinc sulfate can be made from the reaction of sulfuric acid with zinc oxide or zinc.

Give the name of another compound that reacts with sulfuric acid to produce zinc sulfate.

..... [1]

**(f)** A student reacts zinc with excess sulfuric acid.
She obtains 16.1 g of zinc sulfate from 6.5 g of zinc.

**(i)** Calculate the mass of zinc sulfate she would obtain from 26.0 g of zinc.

[1]

**(ii)** Calculate the relative formula mass of zinc sulfate, $ZnSO_4$.

[2]

[Total: 15]

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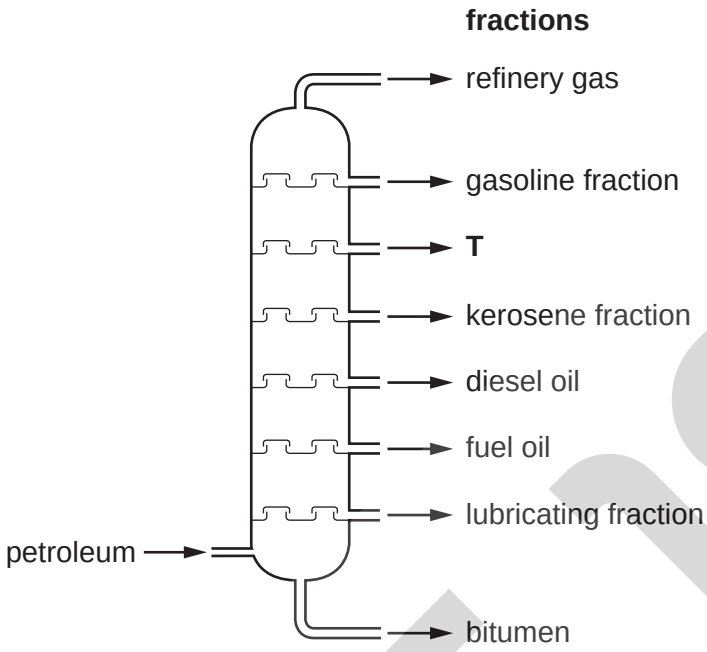
**7** Petroleum is separated into useful fractions by fractional distillation.




**(a)** **(i)** Put an **X** on the diagram to show where the temperature in the column is the highest. [1]

**(ii)** Give the name of the fraction labelled **T**.

..... [1]

**(iii)** The lubricating fraction is used to make lubricants.

Give **one** other use of this fraction.

..... [1]

**(b)** Each fraction contains alkanes.

Which **two** of the following statements are correct?
Tick **two** boxes.

- Alkanes burn to form carbon dioxide and hydrogen. ☐
- Ethene is an alkane with two carbon atoms. ☐
- Alkanes polymerise to form poly(alkanes). ☐
- Alkanes are generally unreactive apart from burning. ☐
- Methane is an alkane present in natural gas. ☐

[2]

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**(c)** Hydrogen can be made by cracking.

**(i)** What is meant by the term *cracking*?

.....

..... [2]

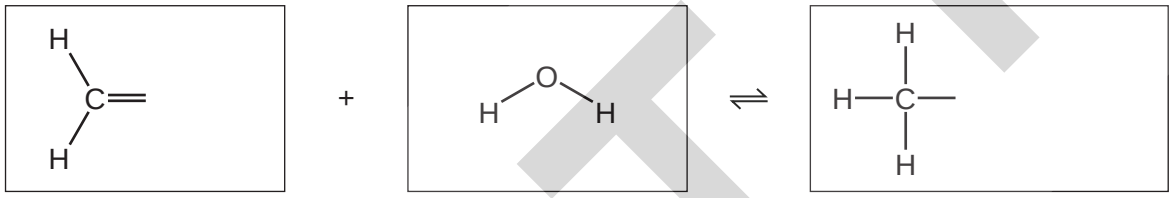
**(ii)** Complete the equation for the cracking of propane.

$$C_3H_8 \rightarrow \text{.....} + H_2$$

[1]

**(d)** Ethanol is formed by the catalytic addition of steam to ethene.

**(i)** Complete the structures of ethene and ethanol in the equation below, showing all atoms and bonds.




ethene ethanol

[2]

**(ii)** What does the symbol ⇌ mean?

..... [1]

[Total: 11]

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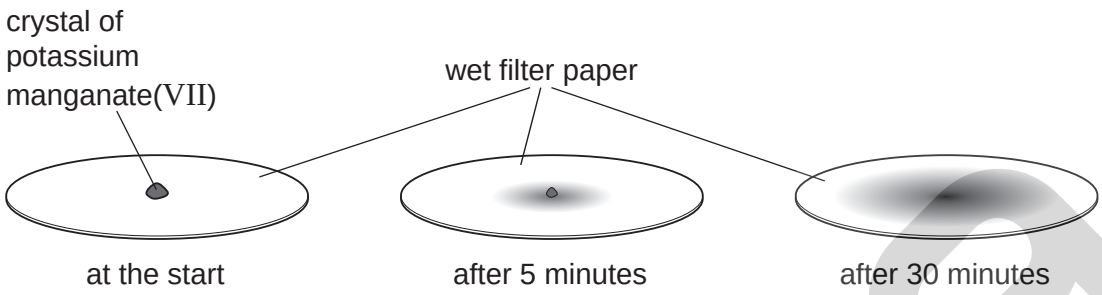
**8** A student placed a crystal of purple potassium manganate(VII) on a filter paper which had been soaked in water.
After 5 minutes, a purple colour had spread out from the crystal.
After 30 minutes, the purple colour had spread further out.




**(a)** Use the kinetic particle theory to explain these observations.

.....

.....

.....

.....

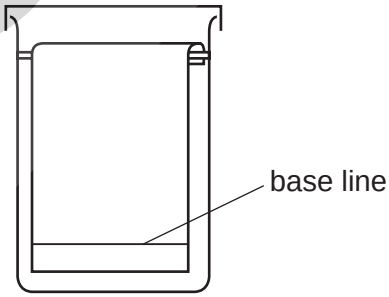
..... [3]

**(b)** Describe the closeness and motion of the particles in a crystal of potassium manganate(VII).

closeness .....

motion .....
[2]

**(c)** Mixtures of dyes can be separated by paper chromatography using the apparatus shown below.




On the diagram above

- draw a line to show the solvent level at the beginning of the experiment,
- put a cross to show where the spot of dye mixture is placed at the beginning of the experiment.

[2]

[Total: 7]

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## DATA SHEET
## The Periodic Table of the Elements

| Group I | Group II | | | | | | | | | | | Group III | Group IV | Group V | Group VI | Group VII | Group 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | 1 **H** Hydrogen 1 | | | | | | | | | | 4 **He** Helium 2 |
| 7 **Li** Lithium 3 | 9 **Be** Beryllium 4 | | | | | | | | | | | 11 **B** Boron 5 | 12 **C** Carbon 6 | 14 **N** Nitrogen 7 | 16 **O** Oxygen 8 | 19 **F** Fluorine 9 | 20 **Ne** Neon 10 |
| 23 **Na** Sodium 11 | 24 **Mg** Magnesium 12 | | | | | | | | | | | 27 **Al** Aluminium 13 | 28 **Si** Silicon 14 | 31 **P** Phosphorus 15 | 32 **S** Sulfur 16 | 35.5 **Cl** Chlorine 17 | 40 **Ar** Argon 18 |
| 39 **K** Potassium 19 | 40 **Ca** Calcium 20 | 45 **Sc** Scandium 21 | 48 **Ti** Titanium 22 | 51 **V** Vanadium 23 | 52 **Cr** Chromium 24 | 55 **Mn** Manganese 25 | 56 **Fe** Iron 26 | 59 **Co** Cobalt 27 | 59 **Ni** Nickel 28 | 64 **Cu** Copper 29 | 65 **Zn** Zinc 30 | 70 **Ga** Gallium 31 | 73 **Ge** Germanium 32 | 75 **As** Arsenic 33 | 79 **Se** Selenium 34 | 80 **Br** Bromine 35 | 84 **Kr** Krypton 36 |
| 85 **Rb** Rubidium 37 | 88 **Sr** Strontium 38 | 89 **Y** Yttrium 39 | 91 **Zr** Zirconium 40 | 93 **Nb** Niobium 41 | 96 **Mo** Molybdenum 42 | **Tc** Technetium 43 | 101 **Ru** Ruthenium 44 | 103 **Rh** Rhodium 45 | 106 **Pd** Palladium 46 | 108 **Ag** Silver 47 | 112 **Cd** Cadmium 48 | 115 **In** Indium 49 | 119 **Sn** Tin 50 | 122 **Sb** Antimony 51 | 128 **Te** Tellurium 52 | 127 **I** Iodine 53 | 131 **Xe** Xenon 54 |
| 133 **Cs** Caesium 55 | 137 **Ba** Barium 56 | 139 **La** Lanthanum 57 * | 178 **Hf** Hafnium 72 | 181 **Ta** Tantalum 73 | 184 **W** Tungsten 74 | 186 **Re** Rhenium 75 | 190 **Os** Osmium 76 | 192 **Ir** Iridium 77 | 195 **Pt** Platinum 78 | 197 **Au** Gold 79 | 201 **Hg** Mercury 80 | 204 **Tl** Thallium 81 | 207 **Pb** Lead 82 | 209 **Bi** Bismuth 83 | **Po** Polonium 84 | **At** Astatine 85 | **Rn** Radon 86 |
| **Fr** Francium 87 | 226 **Ra** Radium 88 | 227 **Ac** Actinium 89 † | | | | | | | | | | | | | | | |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 140 **Ce** Cerium 58 | 141 **Pr** Praseodymium 59 | 144 **Nd** Neodymium 60 | **Pm** Promethium 61 | 150 **Sm** Samarium 62 | 152 **Eu** Europium 63 | 157 **Gd** Gadolinium 64 | 159 **Tb** Terbium 65 | 162 **Dy** Dysprosium 66 | 165 **Ho** Holmium 67 | 167 **Er** Erbium 68 | 169 **Tm** Thulium 69 | 173 **Yb** Ytterbium 70 | 175 **Lu** Lutetium 71 |
| 232 **Th** Thorium 90 | **Pa** Protactinium 91 | 238 **U** Uranium 92 | **Np** Neptunium 93 | **Pu** Plutonium 94 | **Am** Americium 95 | **Cm** Curium 96 | **Bk** Berkelium 97 | **Cf** Californium 98 | **Es** Einsteinium 99 | **Fm** Fermium 100 | **Md** Mendelevium 101 | **No** Nobelium 102 | **Lr** Lawrencium 103 |

*58-71 Lanthanoid series
†90-103 Actinoid series

Key

| a |
|---|
| **X** |
| b |

a = relative atomic mass
**X** = atomic symbol
b = proton (atomic) number

The volume of one mole of any gas is 24 $dm^3$ at room temperature and pressure (r.t.p.).

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